

"RPI_Robot_HAT" is a raspberry PI HAT designed for small/medium size robots. It has an IMU, Dynamixel connectors and audio. Its dimensions fits a raspberry PI zero

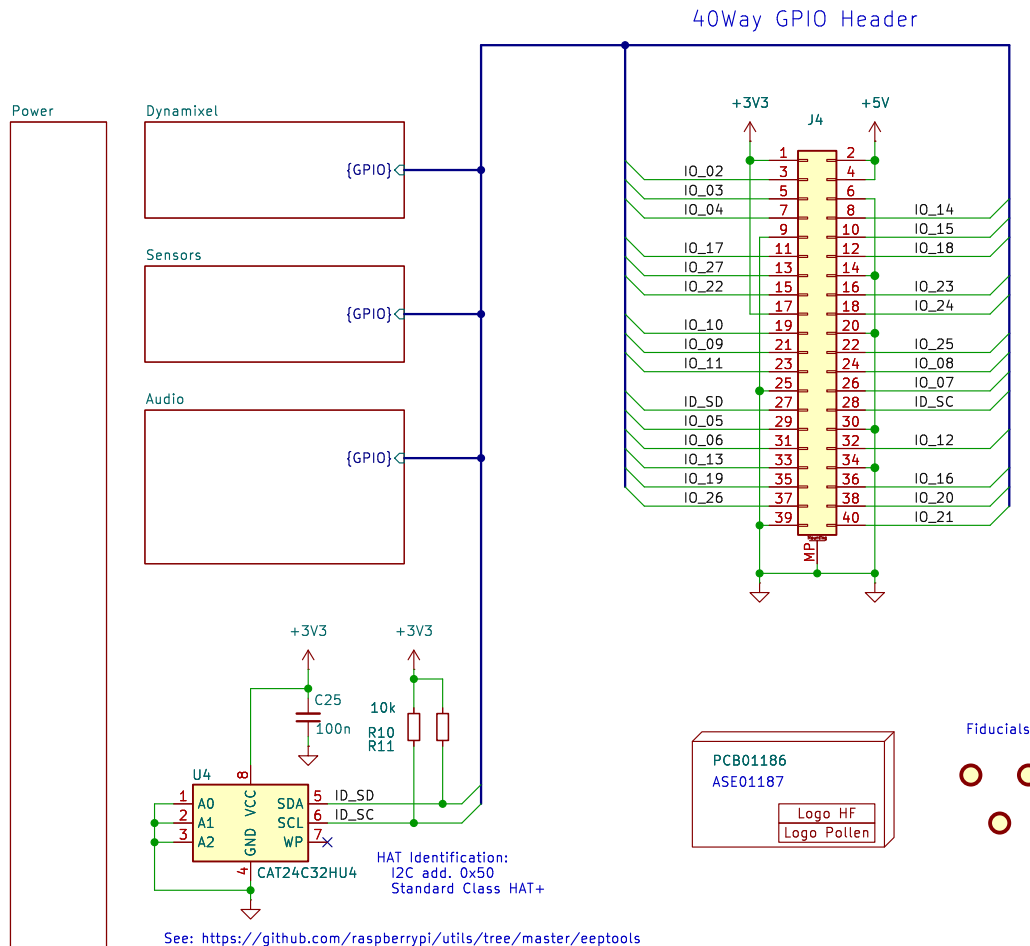
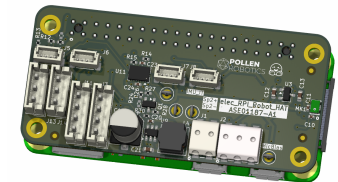
Release	Date	Designer	Check	Comments
A1	06/03/2026	EB	SN	Initial drawing
				ToDo: - HAT+ marking?

Cabled buses on Raspberry Pis

- PI zero 2W:
 - I2C0 (pin 27/28 ID_SD/C): HAT identification (reserved)
 - I2C1 (pin 2/3 SDA/SCL): IMU + Audio + Stemma/QT
- PI 4:
 - same I2C0/1
 - I2C4 (pin 8/9 SDA/SCL): Stemma/QT
 - I2C5 (pin 10/11 SDA/SCL): Stemma/QT
- PI 5:
 - same I2C0/1
 - I2C2 (pin 4/5 SDA/SCL): Stemma/QT

Cabled GPIOs on the 40-way Header:

- GPIO00/01: HAT identification over I2C (reserved)
- GPIO02: IMU + Audio + Stemma/QT - SDA
- GPIO03: IMU + Audio + Stemma/QT - SCL
- GPIO04: Stemma/QT - SDA
- GPIO05: Stemma/QT - SCL
- GPIO06
- GPIO07
- GPIO08: Stemma/QT - SDA
- GPIO09: Stemma/QT - SCL
- GPIO10: Stemma/QT - SDA
- GPIO11: Stemma/QT - SCL
- GPIO12
- GPIO13: Dynamixel - UART_RTS
- GPIO14: Dynamixel - UART_Tx
- GPIO15: Dynamixel - UART_Rx
- GPIO16
- GPIO17
- GPIO18: Audio, PCM Clk
- GPIO19: Audio, PCM FS
- GPIO20: Audio, Din
- GPIO21: Audio, Dout
- GPIO22
- GPIO23
- GPIO24
- GPIO25
- GPIO26
- GPIO27



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Sheet: /
File: elec_RPI_Robot_HAT.kicad_sch

Title: elec RPI Robot HAT

Size: A4 Date: 2026-03-06
KiCad E.D.A. 9.0.7

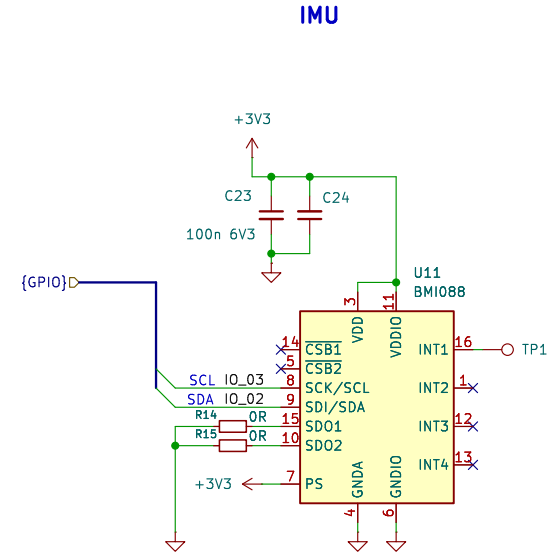
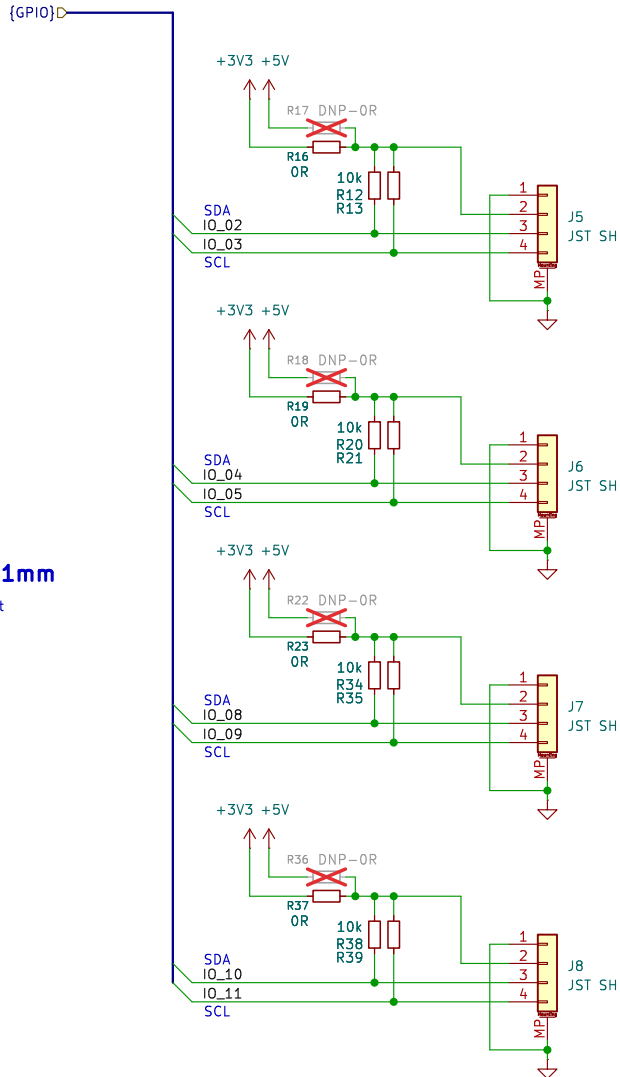
Rev: A1
Id: 1/5



See: https://emanual.robotis.com/docs/en/parts/interface/dynamixel_shield/ as a reference.



Sensors



Pollen Robotics / Hugging Face

Sheet: /Sensors/

File: sensors.kicad_sch

Title: RPI Robot Board – CMx Carrier

Size: A4

Date: 2026-01-05

Rev: A1

KiCad E.D.A. 9.0.7

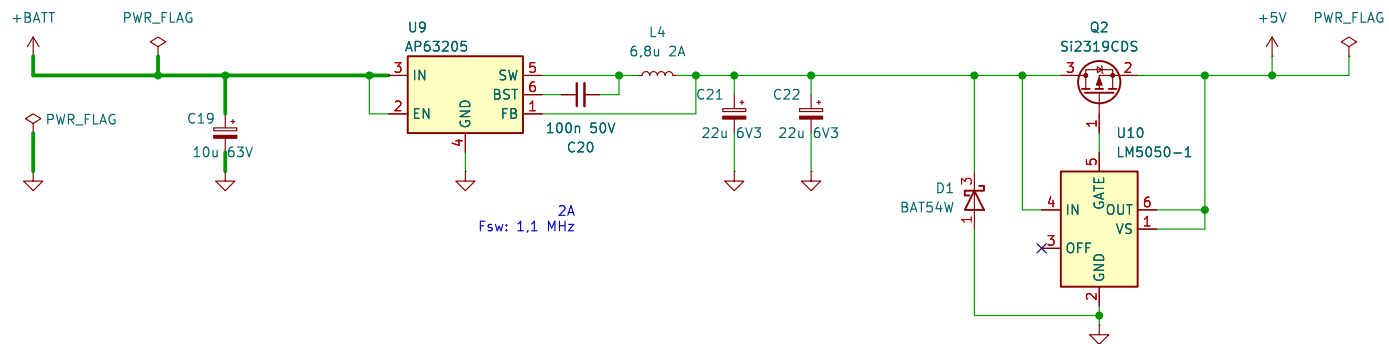
Id: 3/5

Power

24V
(i.e. 7s/25c,
from 21V to 29V)

ToDo:

- 2A DCDC is a bit weak (1,4A for CM4)
- Battery input
- protections (reverse, OV, OC)
- 5V reg for the CM
- Battery voltage output (battery bus)



Model	Idle (W)	Charge max. (W)	"Usage moyen"* (W)
Raspberry Pi 5	2,4 - 3,3	9 - 10	4,0
Raspberry Pi 4 B	2,5 - 3,4	6 - 7,6	3,5
Raspberry Pi Zero 2 W	0,6	1,5	0,8



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Sheet: /Power/
File: power.kicad_sch

Title: RPI Robot Board – CMx Carrier

Size: A4

Date: 2026-01-05

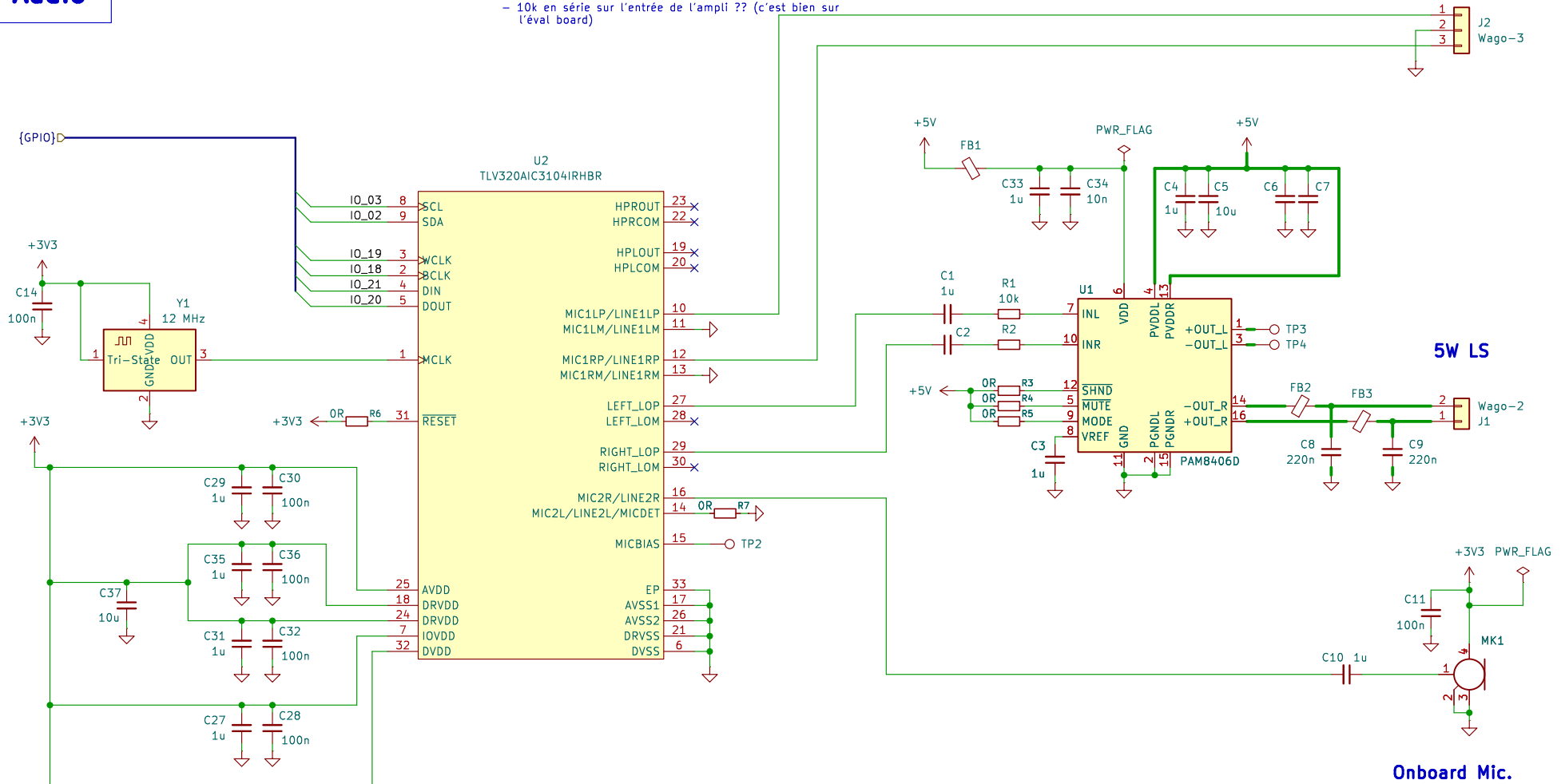
Rev: A1

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Id: 4/5

Audio

ToDo:
- 10k en série sur l'entrée de l'ampli ?? (c'est bien sur l'éval board)



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Sheet: /Audio/

File: audio.kicad_sch

Title: elec RPI Robot HAT

Size: A4

Date: 2026-02-23

KiCad E.D.A. 9.0.7

Rev: A1

Id: 5/5